

SIMATS ENGINEERING

COURSE NAME: FUNDAMENTALS OF DATA SCIENCE

COURSE CODE: DSA0404

CO 3 - ASSESSMENT TOOL 2: Data Interpretation

NAME : H.Ganesh Kumar

REG NO: 192572027

COURSE OUTCOME COVERED

- Apply data preparation and exploratory data analysis techniques, including data summarization, data distribution analysis, outlier treatment, and measures of asymmetry, to preprocess and interpret datasets effectively.
- Analyze datasets using descriptive statistical measures such as mean, variance, sampling methods, covariance, and correlation to identify patterns, relationships, and support data-driven decision-making.

Sl.No 31-40**1. Continuous Distribution — Area Interpretation**

A continuous random variable X is normally distributed with $\mu = 100$, $\sigma = 15$.

(a) What proportion of values lie between 85 and 130? (b) A value of 145 is observed — how many standard deviations away is it, and would you flag it as a potential outlier under the empirical rule?

Solution: Continuous Distribution — Area Interpretation

Given $X \sim N(100, 15^2)$.

(a) Proportion between 85 and 130:

$$z_1 = \frac{85 - 100}{15} = -1.00, z_2 = \frac{130 - 100}{15} = +2.00$$

$$P(-1 < Z < 2) = 0.9772 - 0.1587 = 0.8185$$

Approximately 81.85% of values lie between 85 and 130. (Cross-check with the empirical rule: this region is the union of half the $\pm 1\sigma$ band on the left, 34.13%, and half the $\pm 2\sigma$ band on the right, 47.72% — which sums to 81.85%.)

(b) Evaluating the value 145:

$$z = \frac{145 - 100}{15} = \frac{45}{15} = 3.00$$

The observation lies exactly 3 standard deviations above the mean.

Yes, flag it as a potential outlier. Under the empirical rule, 99.7% of a normal distribution falls within $\pm 3\sigma$, leaving only 0.3% outside — and just 0.135% above $+3\sigma$, roughly 1 in 741 observations. This is the conventional threshold at which a value is treated as anomalous.

One important qualification: "potential outlier" means *investigate*, not *delete*. In a sample of 10,000 observations you would expect about 13 values beyond $+3\sigma$ purely by chance, so a z of 3 is unremarkable in large datasets and highly suspicious in small ones. The correct next step is to verify whether 145 is a recording error, a member of a different population, or a genuine extreme observation worth studying in its own right.

2. Sampling Bias vs Sampling Error

A survey estimating average household expenditure samples only online panel respondents. The sample mean is ₹18,000 vs a known population mean of ₹14,500 (from census data). (a) Is this discrepancy better explained by sampling error or sampling bias? (b) Would increasing the sample size fix this problem? Justify.

Sampling Bias vs Sampling Error

(a) This is sampling bias, not sampling error.

The distinguishing features are direction and magnitude. Sampling error is random: it arises because any given sample differs by chance from the population, and it is equally likely to overshoot or undershoot. Sampling bias is systematic: a flaw in the sampling frame causes the estimate to miss in a consistent direction regardless of sample size.

The discrepancy here is ₹18,000 versus ₹14,500 — an overestimate of 24%, far too large and too directional to be chance. And the mechanism is transparent: an online panel is not a representative frame for household expenditure. It systematically excludes households without reliable internet access, digital literacy, or the leisure to join survey panels — which skews the sample toward wealthier, urban, younger, and better-educated respondents, precisely the groups that spend more. This is textbook coverage bias compounded by self-selection bias, since panel members volunteer rather than being randomly selected.

(b) No — increasing the sample size will not fix this.

This is the single most important distinction in survey methodology. Sample size affects only the *variance* of an estimator, not its *expectation*. Formally, sampling error shrinks as $SE = \sigma/\sqrt{n}$ and vanishes as n grows, but bias is a fixed offset that no amount of data can erode. Surveying 100,000 online panellists instead of 1,000 would simply produce a more precisely wrong answer — the estimate would converge tightly on ₹18,000, which is the mean of the online population, not the mean of all households. The confidence interval would narrow around a value that never contained the truth, creating dangerous false confidence.

What would actually fix it: redesign the sampling frame to cover all households (multi-mode collection including face-to-face and telephone), use probability sampling with known selection probabilities, apply post-stratification weighting to align sample demographics with census margins, or calibrate against known auxiliary totals. Bias is a design problem and requires a design solution.

3. Covariance Sign vs Correlation Strength

Two variable pairs both show positive covariance: Pair 1 has $\text{Cov} = 850$, Pair 2 has $\text{Cov} = 12$. Explain why you cannot conclude Pair 1 has a "stronger relationship" than Pair 2 without additional information, and state what that missing information is.

Solution: Covariance Sign vs Correlation Strength

Why the comparison is invalid. Covariance is defined as

$$\text{Cov}(X' Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

and it carries the product of the two variables' units — rupees $\times$ kilograms, metres $\times$ seconds, and so on. It has no upper or lower bound, so its magnitude confounds two entirely different things: the *strength* of the association and the *scale* of the variables.

A concrete illustration makes the trap obvious. Suppose Pair 1 measures annual income ($\text{SD} \approx 500,000$) against house price ($\text{SD} \approx 3,000,000$) — its covariance could be in the millions while representing a weak link. Meanwhile Pair 2 might measure two standardized test scores ($\text{SD} \approx 3.5$ each), where a covariance of 12 would imply

$$r = \frac{12}{3.5 \times 3.5} = 0.98$$

— an almost perfect relationship. Worse, covariance is not even stable for a *fixed* dataset: convert one variable from rupees to thousands of rupees and the covariance falls by a factor of 1,000 while the underlying relationship is untouched. Any quantity that changes when you change units cannot be a measure of association strength.

The missing information: the standard deviations of X and Y in each pair. With those, you compute the correlation coefficient

$$r = \frac{\text{Cov}(X' Y)}{\sigma_X \sigma_Y}$$

which divides out both sets of units and confines the result to $[-1, +1]$. Only then are the two pairs comparable on a common scale.

The rule to carry forward: covariance's sign is meaningful, its magnitude is not. Both pairs here move in the same direction — that is all you can legitimately say.

4. Estimation — Confidence Interval Reasoning

A sample of $n = 100$ items has mean $= 250$ and $SD = 40$. Construct a 95% confidence interval for the population mean. Then explain, in interpretation (not formula) terms, what this interval does and does NOT tell you about the true population mean.

Solution: Estimation — Confidence Interval Reasoning

Given $n = 100$, $\bar{x} = 250$, $s = 40$.

$$SE = \frac{s}{\sqrt{n}} = \frac{40}{10} = 4$$

$$CI_{95\%} = \bar{x} \pm 1.96 \times SE = 250 \pm 1.96(4) = 250 \pm 7.84$$

$$(242.16, 257.84)$$

(With $n = 100$, the z -value of 1.96 is appropriate; the t -value of 1.984 would widen the interval only marginally, to roughly 242.06–257.94.)

What the interval DOES tell you. It describes the reliability of the *procedure*. If you repeated this study many times — drawing fresh samples of 100 and constructing an interval each time — approximately 95% of those intervals would contain the true population mean μ . It also communicates precision: the margin of error is ± 7.84 , about 3% of the estimate, so the data pin down μ reasonably tightly. And it functions as an implicit hypothesis test: any hypothesized value outside this range (say, $\mu = 260$) would be rejected at the 5% level.

What the interval does NOT tell you.

- It does not mean "there is a 95% probability that μ lies between 242.16 and 257.84." In frequentist statistics μ is a fixed constant, not a random variable. This particular interval either contains it or does not — the probability is 1 or 0, we simply do not know which. The 95% describes the long-run behaviour of the method, not this one result.
- It does not mean 95% of individual items fall in this range. That would be a *prediction interval* and would be far wider, roughly 250 ± 78 . The CI describes uncertainty about the mean, not the spread of the data.
- It does not mean 95% of future sample means will land in this interval — that proportion is closer to 83%.
- It says nothing about bias. If the sample was not randomly drawn, the interval is precise around the wrong target, and the 95% guarantee collapses entirely.

- Values inside the interval are not equally plausible; 250 is considerably better supported than 242.2.

5. Integrated Case Study

A retail company's daily sales data (in ₹'000) for 15 days is:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

- Identify the outlier and justify using both the IQR method and z-score method.
- Compute mean and median with and without the outlier — compare the sensitivity of each.
- Compute skewness with the outlier included — classify the distribution shape.
- Recommend whether the company should remove, cap, or investigate the outlier before reporting monthly performance, and justify your recommendation using the statistical evidence from (a)–(c).

Integrated Case Study

Data (n = 15): 120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125.

Sorted: 117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500. Sum = 2,227.

(a) Identifying the outlier

IQR method. Lower half (positions 1–7) gives Q1 = 120; upper half (positions 9–15) gives Q3 = 128.

Statistic	Value
Q1	120
Q3	128
IQR	8
Lower fence = 120 – 1.5(8)	108
Upper fence = 128 + 1.5(8)	140

500 lies far beyond the upper fence of 140 — it exceeds it by 360, which is 45 times the entire IQR. No other value is outside the fences. Even the 3×IQR "extreme outlier" fence of 152 is comfortably breached.

Z-score method. Mean = 2,227/15 = 148.47. Using $\sum x^2 = 463,255$:

$$\sigma^2 = \frac{463,255}{15} - (148.47)^2 = 30,883.67 - 22,042.36 = 8,841.31, \sigma = 94.03$$

$$z_{500} = \frac{500 - 148.47}{94.03} = 3.74$$

This exceeds the ± 3 threshold, confirming the outlier.

A subtle but revealing detail: for $n = 15$, the maximum attainable z-score is $\sqrt{n-1} = \sqrt{14} = 3.742$. The observation has hit that ceiling exactly. This is the masking effect — the outlier inflates the SD used to judge it, so it partially conceals its own extremity. The IQR method, using only order statistics, is immune to this and delivers the far more emphatic verdict. Both methods agree that 500 is the outlier, but the IQR method reveals the true severity.

(b) Sensitivity of mean vs median

Statistic	With outlier	Without outlier (n = 14)	Change	% change
Mean	148.47	123.36	-25.11	-16.9%
Median	124.00	123.50	-0.50	-0.4%

The contrast is stark. Removing one observation out of fifteen shifts the mean by nearly 17% but the median by four-tenths of one percent — the mean is roughly 40 times more sensitive.

More damning still: the mean of 148.47 exceeds every single one of the other fourteen days, the highest of which is 130. As a summary of daily performance it is not merely imprecise, it is impossible — it describes a sales level the company never once achieved on a normal day. The median of 124, by contrast, sits squarely among the actual observations both before and after removal.

(c) Skewness with the outlier included

Pearson's second coefficient:

$$S_k = \frac{3(\bar{x} - M)}{\sigma} = \frac{3(148.47 - 124)}{94.03} = \frac{73.40}{94.03} = +0.78$$

Moment coefficient of skewness:

$$g = \frac{m_3}{m^{3/2}} = \frac{2,880,225}{831,313} = +3.47$$

Both are positive, so the distribution is strongly right-skewed (positively skewed) with a single extreme value forming a long right tail. The ordering Mean (148.47) > Median (124) > Mode (125) confirms it.

The wide gap between the two coefficients is itself informative. Pearson's version reports a merely "moderate" +0.78 because it is bounded by ± 3 and heavily damped by the inflated SD in its denominator — the same masking effect seen in part (a). The moment coefficient, which cubes deviations and therefore amplifies the distant point, returns +3.47, essentially at the theoretical

maximum of $\frac{n-2}{\sqrt{n-1}} = 3.474$ for $n = 15$. When a skewness statistic saturates at its own upper bound,

the asymmetry is being produced by a single observation, not by the shape of the underlying process.

(d) Recommendation: investigate first, then decide

Do not remove or cap the value before establishing what it is. Deletion and Winsorization are both statistical treatments for a problem that has not yet been diagnosed, and the two possible diagnoses demand opposite responses.

The statistical evidence establishes three things. First, from (a), the value is unambiguously anomalous by two independent methods and is not a borderline case. Second, from (b), the fourteen remaining days are extraordinarily tight — spanning just 117 to 130, a range of 13 with an IQR of 8. This is a highly stable, low-variance business process, which makes a jump to nearly four times the typical daily level structurally implausible as ordinary fluctuation. Third, from (c), the skewness statistics are pinned at their theoretical maxima, confirming a single-point artefact rather than a genuinely skewed sales distribution.

Investigation should test two specific hypotheses:

- A data-entry error. The tight clustering and the suspicious roundness of "500" point this way. Common mechanisms include a transposition, a decimal misplacement (₹50.0 thousand entered as 500), or a duplicate batch entry. Cross-check against POS logs, invoices, and bank deposits for that date.
- A genuine business event. A festival, a corporate bulk order, a clearance sale, or a one-off wholesale transaction can legitimately produce a 4× day. Check the calendar date, transaction counts, and whether the day is identifiable as special.

The resulting decision rule:

Finding	Action	Rationale
Confirmed data-entry error	Correct to the true value; if unrecoverable, treat as missing	A wrong number carries no information and must not be preserved
Genuine one-off event	Retain, but report it separately with annotation	Deleting real revenue misstates performance and destroys business insight
Cause undeterminable	Report median and trimmed mean as headline figures; disclose the anomaly	Transparency over silent adjustment

For the monthly performance report specifically, lead with the median of 124 or a trimmed mean rather than the raw mean of 148.47, since the latter overstates typical daily sales by over 20% and would corrupt any downstream inventory planning, staffing, or forecasting built on it. Whatever treatment is chosen, document it explicitly. Silently deleting an observation that inflates results is an audit and governance failure regardless of whether the statistics justify it — and if 500 turns out to be real, it is the single most commercially interesting data point of the month and deserves analysis rather than erasure.